Runbook: MCO User Lock / Restore — PRODUCTION

Item	Value
Script	`Set-McoUserLock-PROD.ps1`
Purpose	Temporarily disable all active MCO users (option 1) and restore them to their original values afterwards (option 2)
Database	Oracle 11g, host `172.20.3.16`, port `1526`, SID `MCOPROD`
Table	`MCO001.UTILISATEUR`
Excluded users (never changed)	`PATKE01`, `CICAM01`, `KHARI01`, `TCHAR02`
Credential file	`MCOPROD.cred` (next to the script)
Backup file	`UTILISATEUR_locked_users_PROD.csv` (next to the script)
Runtime	Windows PowerShell 5.1
Environment	**PRODUCTION** — DEV equivalent: `Set-McoUserLock.ps1` on `MCODEV`

> **Production change.** Run only with an approved change ticket, in the approved window, after the same steps have been tested on MCODEV. Every save requires answering **Y** *and* typing `MCOPROD`.

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1. What the script does

1.1 Option 1 — Disable active users

An **active user** is one where `NBRE_TENTATIVE = 0` **and** `COMPTE_VEROUILLE = 0`.

Step	Action
1	Checks that the backup file does **not** already exist. If it does, the script stops, so original values are never overwritten
2	Connects to `172.20.3.16:1526/MCOPROD` and starts a transaction
3	Selects and locks (`FOR UPDATE`) every active user except the excluded codes
4	Lists the users and asks **Disable N user(s) listed above?** (default **No**)
5	On **Yes**: asks you to type `MCOPROD` (capitals). Anything else cancels with no change
6	Writes the original values to the backup CSV **before** saving any change

Step	Action
7	Sets `NBRE_TENTATIVE = 3` and `COMPTE_VEROUILLE = 1` for each listed user
8	Commits only if the number of rows updated matches the number listed; otherwise rolls back and deletes the backup

```
-- Selection
SELECT CODE_UTILISATEUR, NOM_UTILISATEUR, PRENOM_UTILISATEUR, NBRE_TENTATIVE, COMPTE_VEROUILLE
  FROM MCO001.UTILISATEUR
 WHERE NBRE_TENTATIVE = 0 AND COMPTE_VEROUILLE = 0
   AND UPPER(TRIM(CODE_UTILISATEUR)) NOT IN ('PATKE01', 'CICAM01', 'KHARI01', 'TCHAR02')
   FOR UPDATE;

-- Update (per user)
UPDATE MCO001.UTILISATEUR
   SET NBRE_TENTATIVE = 3, COMPTE_VEROUILLE = 1
 WHERE CODE_UTILISATEUR = :code AND NBRE_TENTATIVE = 0 AND COMPTE_VEROUILLE = 0;
```

1.2 Option 2 — Restore users

Step	Action
1	Checks that the backup file exists
2	Lists the users in the backup and asks **Restore N user(s) listed above to their original values?** (default **No**), then asks you to type `MCOPROD`
3	For each user, sets `NBRE_TENTATIVE` and `COMPTE_VEROUILLE` back to their original values, **only if** they still hold `3` and `1`
4	Users changed by someone else since the lock are **skipped** and listed in a warning
5	Commits all restores in one transaction
6	Renames the backup to `UTILISATEUR_locked_users_PROD.restored_yyyyMMdd_HHmmss.csv` so it can't be applied twice

```
UPDATE MCO001.UTILISATEUR
   SET NBRE_TENTATIVE = :origN, COMPTE_VEROUILLE = :origC
 WHERE CODE_UTILISATEUR = :code AND NBRE_TENTATIVE = 3 AND COMPTE_VEROUILLE = 1;
```

1.3 Backup file format

CSV with a header row, UTF-8. One row per user changed by option 1.

Column	Content
`CODE_UTILISATEUR`	User code
`NOM_UTILISATEUR` / `PRENOM_UTILISATEUR`	Last name / first name
`ORIG_NBRE_TENTATIVE` / `NEW_NBRE_TENTATIVE`	Value before (`0`) / after (`3`)

Column	Content
`NBRE_TENTATIVE_CHG`	`Y` if the column was changed
`ORIG_COMPTE_VEROUILLE` / `NEW_COMPTE_VEROUILLE`	Value before (`0`) / after (`1`)
`COMPTE_VEROUILLE_CHG`	`Y` if the column was changed
`CHANGED_AT`	Timestamp of the lock run

> **Do not edit, move or delete the backup file between option 1 and option 2.** It is the only record of the original values.

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2. Prerequisites

#	Requirement	How to check
1	Windows PowerShell 5.1	`$PSVersionTable.PSVersion`
2	Network access to `172.20.3.16` on port `1526`	`Test-NetConnection 172.20.3.16 -Port 1526`, which should return `TcpTestSucceeded : True`
3	Oracle account with `SELECT` and `UPDATE` on `MCO001.UTILISATEUR`	The same login can run an `UPDATE` in SQL Developer
4	`Oracle.ManagedDataAccess.dll` **version 19.18.0** in the same folder as the script	See section 3.1

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3. One-time setup

3.1 Install the Oracle driver (19.18.0)

Skip this step if the DLL is already in the folder (e.g. from the AGPH extract setup).

```
cd $env:USERPROFILE\Downloads
[Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12
Invoke-WebRequest https://www.nuget.org/api/v2/package/Oracle.ManagedDataAccess/19.18.0 -OutFile odp.zip
Expand-Archive odp.zip -DestinationPath odp -Force
Copy-Item .\odp\lib\net40\Oracle.ManagedDataAccess.dll .
Unblock-File .\Oracle.ManagedDataAccess.dll
Unblock-File .\Set-McoUserLock-PROD.ps1
```

> **Use 19.18.0, not 23.x.** Version 23.x needs extra .NET files that aren't on a standard Windows PC.

3.2 Save the Oracle credential

The script handles this itself. On the first run it asks for the Oracle user name and password and saves them to `MCOPROD.cred`. Later runs load the file without prompting.

Rule	Reason
Run the script as the Windows user who will use it from now on	The file is encrypted with Windows DPAPI: only that user, on that machine, can decrypt it
Run with `-ResetCred` whenever the Oracle password changes	Otherwise you get `ORA-01017`
Never copy the `.cred` file to another PC or user	It won't decrypt there

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4. Run procedure

4.1 Pre-checks (before option 1)

Check	Action
Change approved	Approved change ticket for MCOPROD; you are inside the approved window
Tested in DEV	The same run (option 1 then option 2) completed successfully on MCODEV
Rollback ready	Option 2 is the rollback. Know where the backup CSV will be written (script folder)
No leftover backup	`Test-Path .\UTILISATEUR_locked_users_PROD.csv` must return `False`
Expected volume	In SQL Developer: `SELECT COUNT(*) FROM MCO001.UTILISATEUR WHERE NBRE_TENTATIVE = 0 AND COMPTE_VEROUILLE = 0 AND CODE_UTILISATEUR NOT IN ('PATKE01','CICAM01','KHARI01','TCHAR02');` — note the number
Users informed	Users have been told they will be unable to log in

4.2 Option 1 — Disable active users

```
cd $env:USERPROFILE\Downloads
.\Set-McoUserLock-PROD.ps1 -Option 1
```

Expected output:

```
Using credentials for MCO001 from C:\Users\<you>\Downloads\MCOPROD.cred
Connected to PRODUCTION 172.20.3.16:1526/MCOPROD

CODE_UTILISATEUR NOM_UTILISATEUR PRENOM_UTILISATEUR ORIG_NBRE_TENTATIVE ORIG_COMPTE_VEROUILLE
---------------- -------------- ------------------ -------------------- ---------------------
ABCDE01          ...            ...                0                    0
...

Confirm
Disable <N> user(s) listed above?
[Y] Yes  [N] No  [?] Help (default is "N"): Y
*** PRODUCTION: 172.20.3.16:1526/MCOPROD ***
Type MCOPROD to confirm: MCOPROD
Disabled <N> user(s). Originals saved to C:\Users\
<you>\Downloads\UTILISATEUR_locked_users_PROD.csv
```

> `<N>` must match the pre-check count in 4.1. If it doesn't, answer **N** (or type anything other than `MCOPROD`) and investigate. Keep the backup CSV safe until option 2 has run — it is the only record of the original values.

4.3 Option 2 — Restore users

```
cd $env:USERPROFILE\Downloads
.\Set-McoUserLock-PROD.ps1 -Option 2
```

Expected output:

```
Using credentials for MCO001 from C:\Users\<you>\Downloads\MCOPROD.cred
Connected to PRODUCTION 172.20.3.16:1526/MCOPROD
...
Restore <N> user(s) listed above to their original values?
[Y] Yes  [N] No  [?] Help (default is "N"): Y
*** PRODUCTION: 172.20.3.16:1526/MCOPROD ***
Type MCOPROD to confirm: MCOPROD
Restored <N> user(s).
Backup archived to C:\Users\
<you>\Downloads\UTILISATEUR_locked_users_PROD.restored_20261006_153012.csv
```

If some users were changed after the lock, you also see:

```
WARNING: Skipped 2 user(s) - not found or values changed since lock: ABCDE01, FGHIJ02
```

Review those users manually (section 5).

4.4 Interactive menu

Running the script with no `-Option` shows a menu:

```
  *** PRODUCTION ***
  MCO001.UTILISATEUR on 172.20.3.16:1526/MCOPROD
  1) Disable all active users (except PATKE01, CICAM01, KHARI01, TCHAR02)
  2) Restore users from C:\Users\<you>\Downloads\UTILISATEUR_locked_users_PROD.csv

Select option (1/2):
```

4.5 Parameters

Parameter	Default	Use
`-Option`	*(menu)*	`1` = disable, `2` = restore
`-CredFile`	`<script folder>\MCOPROD.cred`	Saved credential file
`-ResetCred`	Off	Prompt again and overwrite the `.cred` file
`-UserName`	*(prompt)*	Oracle user name used when creating the `.cred` file
`-Credential`	*(from .cred)*	Pass a credential directly; bypasses the `.cred` file
`-BackupFile`	`<script folder>\UTILISATEUR_locked_users_PROD.csv`	Where option 1 writes and option 2 reads the original values
`-ExcludeUsers`	`PATKE01`, `CICAM01`, `KHARI01`, `TCHAR02`	User codes never changed by option 1
`-DbHost`	`172.20.3.16`	Database host
`-Port`	`1526`	Listener port
`-Sid`	`MCOPROD`	SID, or the service name when used with `-UseServiceName`
`-UseServiceName`	Off	Connect with `SERVICE_NAME` instead of `SID`
`-Schema`	`MCO001`	Schema owning `UTILISATEUR`
`-DllPath`	`<script folder>\Oracle.ManagedDataAccess.dll`	Driver location

5. Post-run verification

After	Check (SQL Developer)	Pass criteria
Option 1	`SELECT COUNT(*) FROM MCO001.UTILISATEUR WHERE NBRE_TENTATIVE = 0 AND COMPTE_VEROUILLE = 0;`	Only the excluded users remain (≤ 4)
Option 1	`SELECT CODE_UTILISATEUR, NBRE_TENTATIVE, COMPTE_VEROUILLE FROM MCO001.UTILISATEUR WHERE CODE_UTILISATEUR IN ('PATKE01','CICAM01','KHARI01','TCHAR02');`	Unchanged

After	Check (SQL Developer)	Pass criteria
Option 1	Backup file	Exists; row count (excluding header) equals the disabled count
Option 2	Same `COUNT(*)` as option 1	Equals the pre-check count from 4.1 plus the excluded active users
Option 2	Skipped users (if any)	Each one reviewed and fixed manually or confirmed as intended
Option 2	Backup file	Renamed to `*.restored_<timestamp>.csv`

6. Exit codes

Code	Meaning
`0`	Finished: changes committed, cancelled at the prompt, or nothing to do
`1`	Failed. The error is printed (e.g. `Failed: ORA-...`). All changes in that run are rolled back

7. Troubleshooting

Symptom	Cause	Fix
`Backup file '...' already exists` (option 1)	A previous lock has not been restored	Run option 2 first, or use `-BackupFile` with a new name if the old file is no longer needed
`Backup file '...' not found` (option 2)	Wrong folder, or already restored	Run from the folder holding the CSV, or pass `-BackupFile`. Check for a `*.restored_*.csv`
`Cancelled. No changes made.` after typing the name	Typed text was not exactly `MCOPROD` (case-sensitive)	Run again and type `MCOPROD` in capitals
Script waits after `Connected to ...`	Another session holds uncommitted changes on `UTILISATEUR` rows	Ask the other session (or KYNDRYL Oracle Database Support) to commit or roll back; or press Ctrl+C and retry later
`Skipped N user(s) ...` (option 2)	Those users were changed after the lock (e.g. reset by support)	Check each user in SQL Developer and fix manually if needed
`Expected N rows updated, got M. Rolled back.`	Rows changed between select and update	Re-run option 1; nothing was saved
`Cannot read '...MCOPROD.cred'` / `Key not valid for use in specified state`	`.cred` created by another Windows user or PC	Run with `-ResetCred`
`ORA-01017: invalid username/password`	Wrong or changed password	Run with `-ResetCred`

Symptom	Cause	Fix
`ORA-28000: account is locked`	Oracle account locked	Ask KYNDRYL Oracle Database Support to unlock it
`ORA-00942: table or view does not exist`	No grant on `MCO001.UTILISATEUR`, or wrong schema	Ask KYNDRYL Oracle Database Support for `SELECT` / `UPDATE`; check `-Schema`
`ORA-01031: insufficient privileges`	`SELECT` only, no `UPDATE`	Ask KYNDRYL Oracle Database Support for `UPDATE` on `MCO001.UTILISATEUR`
`ORA-12505: listener does not currently know of SID`	`MCOPROD` is a service name, not a SID	Add `-UseServiceName`
`ORA-12170` / `ORA-12541` / timeout	Network or firewall	`Test-NetConnection 172.20.3.16 -Port 1526`; check VPN; if blocked, submit a Firewall Request via ServiceNow
`Oracle.ManagedDataAccess.dll not found`	Driver missing	Section 3.1, or pass `-DllPath`
`... running scripts is disabled on this system`	Execution policy / downloaded file	`Unblock-File .\Set-McoUserLock-PROD.ps1`, or `powershell.exe -ExecutionPolicy Bypass -File .\Set-McoUserLock-PROD.ps1`

8. Security and housekeeping

Item	Guidance
Credentials	Only in the DPAPI-encrypted `MCOPROD.cred`; never in the script, command history or email
Backup CSV	Production data: contains user codes and names. Attach a copy to the change ticket as evidence. Keep it on a bank-managed drive; never commit it to Git (it is in `.gitignore`)
Restored backups	Keep `*.restored_*.csv` as the audit trail per retention policy, then delete
Excluded users	Change only with approval; pass `-ExcludeUsers` rather than editing the script
Script changes	Keep under version control (`ksolo13/fscprd`); test on MCODEV (`Set-McoUserLock.ps1`) before running in PROD

9. Contacts and escalation

Issue	Contact
Database access, grants, row locks	KYNDRYL — Oracle Database Support
Network or firewall to `172.20.3.16:1526`	Please submit a Firewall Request via ServiceNow
MCO application / user impact	Kevin Paterson
Script defects	Kevin Paterson

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10. Change log

Date	Change
2026-10-06	Initial PROD version (`Set-McoUserLock-PROD.ps1`), from the tested MCODEV script: MCOPROD defaults, separate `.cred` and backup files, typed `MCOPROD` confirmation, red PRODUCTION banner
2026-10-29	`TCHAR02` added to the excluded users